AI-Based Fading Channel Type Classification

Project Report

Abstract This project classifies wireless fading channels as Rayleigh or Rician using machine learning. Complex channel samples are simulated, corrupted by additive white Gaussian noise (AWGN) at different signal-to-noise ratios (SNRs), converted into statistical and signal-based features, and classified using SVM, Random Forest and MLP. Accuracy, macro-F1, confusion matrices and SNR-wise results are used for evaluation.

1. Introduction Wireless signals travel through multiple paths because of reflection, diffraction and scattering. The combination of these paths causes fading. Rayleigh fading models a channel without a dominant line-of-sight (LOS) component. Rician fading models a channel with a dominant LOS component plus scattered components. This project uses AI to distinguish these two channel types.

2. Objectives - Generate Rayleigh and Rician channel data. - Add AWGN at multiple SNR levels. - Extract statistical and signal-based features. - Train SVM, Random Forest and MLP classifiers. - Evaluate performance using accuracy, macro-F1 and confusion matrices.

3. Theory **Rayleigh fading:** The complex channel is modelled as many scattered components with no dominant LOS path. Its envelope follows a Rayleigh distribution under the standard assumptions.

Rician fading: A dominant LOS component is combined with diffuse scattered components. The K-factor is the ratio of LOS power to diffuse power. The simulation uses K = 6 dB.

AWGN and SNR: SNR(dB) = 10 log10(signal power/noise power). Lower SNR means noise is stronger relative to the signal. Tested levels are [-5, 0, 5, 10, 15, 20, 25] dB.

4. Dataset The dataset contains 3,500 balanced synthetic channel snapshots, each with 256 complex IQ samples. Samples are generated at seven SNR levels from -5 dB to 25 dB. Average channel power is normalized before adding complex AWGN. The raw dataset is in `data/channel_iq_dataset.npz`; features and labels are in `data/channel_features.csv`.

5. Feature Extraction The 21 features include envelope mean, standard deviation, variance, skewness, kurtosis, median, percentiles, RMS, crest factor and coefficient of variation; power mean, standard deviation and kurtosis; phase standard deviation and phase-difference variation; real/imaginary spread, I/Q covariance and autocorrelation magnitudes at lags 1 and 2.

6. Classifiers - **SVM: RBF kernel with standardized features. - **Random Forest:** ensemble of 250 decision trees. - **MLP:** feed-forward neural network with hidden layers of 64 and 32 neurons. A stratified 75%/25% train/test split is used. Numeric class labels are used internally for compatibility with machine-learning libraries.**

7. Evaluation Metrics Accuracy is the fraction of correct predictions. Macro-F1 averages the F1-score of both classes equally. A confusion matrix counts correct and incorrect predictions. SNR-wise evaluation calculates accuracy and macro-F1 separately at each noise level.

8. Results ### Overall model comparison model overall_accuracy overall_macro_f1 SVM 1.000000 1.000000 Random Forest 0.997714 0.997714 MLP 0.997714 0.997714

Best overall accuracy in this run: **SVM**, accuracy = **1.000**, macro-F1 = **1.000**.

Performance by SNR model snr_db accuracy macro_f1 SVM -5 1.000000 1.000000 SVM 0 1.000000 1.000000 SVM 5 1.000000 1.000000 SVM 10 1.000000 1.000000 SVM 15 1.000000 1.000000 SVM 20 1.000000 1.000000 SVM 25 1.000000 1.000000 Random Forest -5 0.981982 0.981910 Random Forest 0 1.000000 1.000000 Random Forest 5 1.000000 1.000000 Random Forest 10 1.000000 1.000000 Random Forest 15 1.000000 1.000000 Random Forest 20 1.000000 1.000000 Random Forest 25 1.000000 1.000000 MLP -5 0.981982 0.981945 MLP 0 1.000000 1.000000 MLP 5 1.000000 1.000000 MLP 10 1.000000 1.000000 MLP 15 1.000000 1.000000 MLP 20 1.000000 1.000000 MLP 25 1.000000 1.000000

See the included accuracy and macro-F1 plots, plus individual confusion matrices, for visual evaluation.

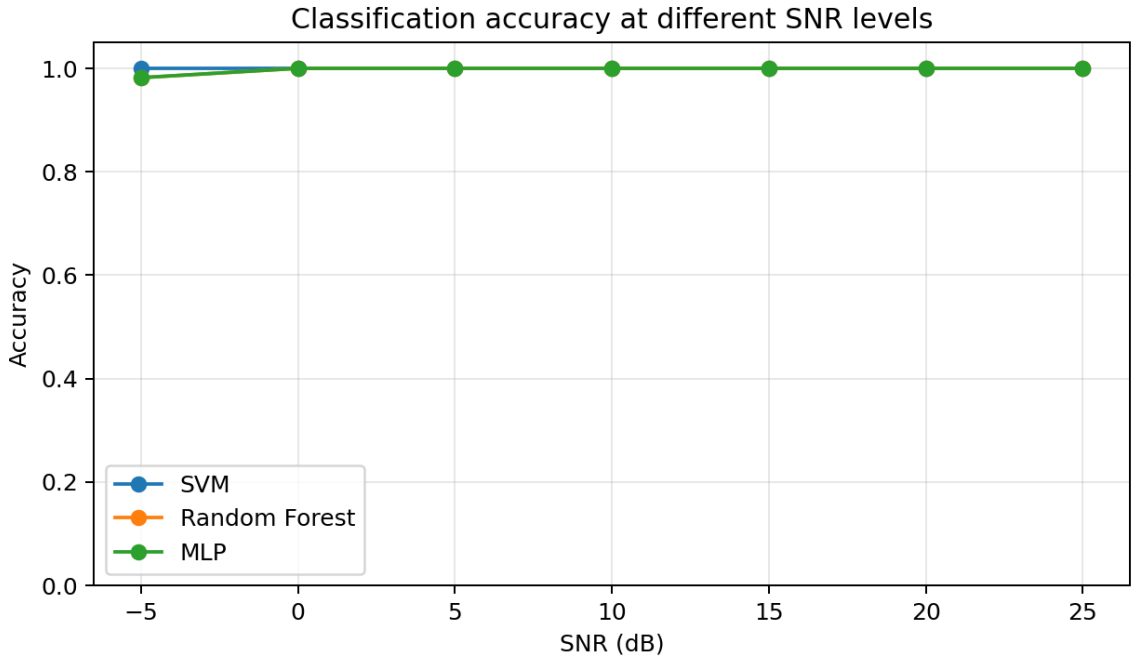
9. Conclusion The project implements the complete workflow of channel generation, feature extraction, machine-learning classification and performance evaluation. The comparison shows how SNR affects the ability to distinguish Rayleigh and Rician fading.

10. Limitations and Future Work This is a simulation-based educational project, not a measurement campaign. It uses two channel types, a fixed

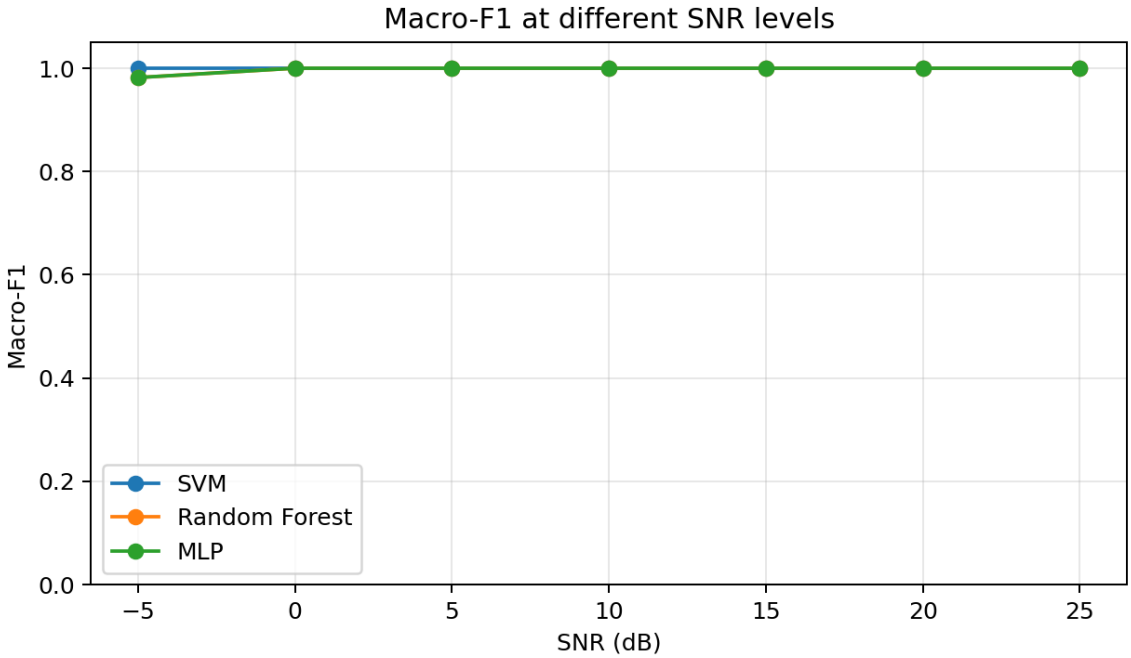
Rician K-factor and a simplified independent-sample model. Future work can use measured IQ data, multiple K-factors, correlated time-varying channels, frequency-selective channels and additional fading distributions such as Nakagami-m.

11. Reproducibility Install dependencies with `pip install -r requirements.txt`, then run `python run_project.py`. The source code regenerates the dataset, extracts features and trains/evaluates the classifiers.

Accuracy versus SNR



Macro-F1 versus SNR


Macro-F1 at different SNR levels
Macro-F1
SNR (dB)
1.0
0.8
0.6
0.4
0.2
0.0
−5
0
5
10
15
20
25
SVM
Random Forest
MLP